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Portfolio Summary

Rev 1, 2026-11-05

About

I'm a second-year Civil Engineering student at Toronto Metropolitan University (BEng, expected 2029), focused on structural engineering. I've worked on site with a design-build firm, tying rebar, building formwork and placing concrete, and as a project coordinator on a mixed-use development of two multi-storey buildings with about 182 residential and seniors-living units, coordinating five consulting disciplines (structural, civil, geotechnical, planning, landscape).

Outside class I build verification packages: I check a structure by hand, in a spreadsheet and in analysis software, chase down every difference between them, and issue drawings and models under revision control. My first package has been revised through three rounds of P.Eng. review. Tools: Robot Structural Analysis, Mathcad Prime, Excel, AutoCAD and Revit (Autodesk Certified User).

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Full documents (calculations, models, drawings and the complete literature review) are in the full portfolio at abbassaleh.com.

All projects are self-directed student exercises and are not for construction.

AS-P1-RPT-001 Technical Brief

Rev 1

Maintenance Platform Verification

2026-10-21

Prepared by: Abbas Saleh Reviewed by: S. Al-Zubaidi, P.Eng. (comments received 2026-10-18; addressed in R1)

Student exercise. Not for construction.

The gravity and lateral framing of a 6.0 m × 3.0 m steel maintenance platform was verified by three independent methods, and the review comments received on R0 were addressed in this revision.

PURPOSE AND SCOPE

The purpose of this brief is to summarize the verification of a steel maintenance platform, which was carried out as a self-directed student exercise to check the member forces, stresses and deflections of the framing by three separate methods and to record the discrepancies found between them. In R0, the checks were limited to elastic strength of materials, which covered the bending stress, shear stress and deflection of the beams and joists and the axial stress and Euler buckling of the posts. In R1, the bracing was then remodelled as tension-only, the lateral loads were checked by hand (CAL-003), and the members were checked to CSA S16 (CAL-004). Apart from the brace end bolt, connections were not designed, and for this reason the work is not intended for construction.

DESIGN BASIS

Item	Basis
Geometry	6.0 m × 3.0 m plan, top of steel 3000 mm
B1 main beams	W310×39, simply supported, 6.0 m span
J1 joists	W150×22 at 1.5 m (5 No.), 3.0 m span
C1 posts	HSS127×127×6.4, 3.0 m, pinned, K = 1.0
BR1 bracing	L64×64×6.4 X-bracing, all four sides, tension-only, bolted at the intersection
Dead load	0.5 kPa grating + self-weight
Live load	4.8 kPa adopted. The minimum for equipment areas and service rooms is 3.6 kPa (NBC 2020, Div. B, Table 4.1.5.3, as adopted by O. Reg. 163/24), and 4.8 kPa was selected to allow for components and tools laid down on the deck during maintenance (cf. storage areas, 4.8 kPa).
Equipment load	10 kN at midspan of each B1 (assumed)
Notional loads	0.005 × 187.32 kN total ULS gravity = 0.94 kN, applied as 0.234 kN at each post top, in X and Y separately (CSA S16, Clause 8.4.1)
Guard load	0.75 kN/m horizontal, outward on one edge at a time: short edge (X, 3 m) and long edge (Y, 6 m) (NBC 2020, Div. B, Article 4.1.5.14)
Load combinations	ULS 1.25D + 1.5L; ULS with lateral 1.25D + 1.5L + 1.5G + 1.0N, with full floor live load combined with guard load (conservative, for review); SLS D + L and D + L + G
Deflection limit	L/360 under D + L adopted, which is more stringent than the L/300 recommended for floors of industrial buildings in CSA S16, Annex D. For B1, D + L gives 10.72 mm (L/560) and live load only gives 9.43 mm (L/636). Checking D + L against a live load limit is conservative.
Lateral support	B1 compression flange braced by J1 at 1.5 m (L_u = 1.5 m); J1 taken as unbraced over 3.0 m, with no restraint assumed from the grating; ω_2 = 1.0
Material	CSA G40.21 350W (assumed), E = 200 000 MPa; HSS taken as Class C (n = 1.34)
Bolts	Assumed 1 M20 A325 per brace end, threads intercepted, 22 mm hole; F_u = 825 MPa (bolt), 450 MPa (350W plate)

Section properties were taken from the CISC section database (beamdimensions.com), and loads marked as assumed were not taken from a cited source.

VERIFICATION METHOD

The framing was checked by three methods which were set up independently of one another, in order that an error in any one method would be shown as a disagreement with the other two:

1. Hand calculation in Mathcad Prime (AS-P1-CAL-001), carried out member by member from joist to beam to post.
2. Parametric spreadsheet in Excel (AS-P1-CAL-002), in which every input was given as a named cell and a CHECK tab was used to compare all three methods, a difference of 2% or less being taken as a pass.
3. 3D frame model in Autodesk Robot Structural Analysis (AS-P1-MOD-001), with member self-weight included and joist end releases set to free both bending rotations while torsion was restrained.

Additionally, the framing was drawn in AutoCAD (S-101 and S-201) and modelled in Revit (AS-P1-MOD-002, sheet S-301), with the Revit schedules being used for the material take-off and as a further check of the modelled sections. In R1, the brace and post forces and the frame displacements under the lateral load cases were also checked by hand in AS-P1-CAL-003.

RESULTS

All beam and joist results were found to agree within 0.1%, and the only difference above that was the post axial load, which is explained under Findings.

Quantity	Unit	Mathcad	Excel	Robot	Robot vs hand	Limit
J1 factored moment M_f	kN·m	13.51	13.52	13.51	0.0%	
J1 factored shear V_f	kN	18.02	18.02	18.02	0.0%	
J1 service deflection	mm	3.56	3.56	3.56	0.0%	L/360 = 8.33
B1 factored moment M_f	kN·m	78.70	78.70	78.69	0.0%	
B1 factored shear V_f	kN	35.96	35.96	35.95	0.0%	
B1 service deflection	mm	10.72	10.72	10.72	0.0%	L/360 = 16.67
C1 factored axial load	kN	46.03	46.03	46.83	+1.7%	
C1 Euler buckling load P_{cr}	kN	1546	1546	not run		

The elastic stresses obtained by hand were 143 MPa bending and 20 MPa shear in B1, 85 MPa bending and 20 MPa shear in J1, and 16 MPa axial in C1, with the factored post load being about 3% of its Euler load. These values are below F_y = 350 MPa, and the member resistances to CSA S16 are given under R1 revisions. The small Mathcad and Excel differences in J1 are attributed to g being taken as 9.807 and 9.81 m/s^2 respectively.

R1 REVISIONS

Tension-only bracing. The X-bracing was modelled as tension-only in MOD-001 R1. Under gravity load all of the braces are placed in compression, which makes them inactive, and because of this the full gravity load is carried by the posts. The C1 member force from Robot was then 46.55 kN at the post base, which is within 1.1% of the 46.03 kN hand value. The support reaction of 46.83 kN is higher, since the brace self-weight is lumped at the base nodes.

Lateral loads. The notional loads and the guard load were applied as separate lateral load cases in X and Y. Since the braces act in tension only, one brace in each X is active for a given load direction. The brace and post forces were then checked by hand in CAL-003, and these agreed with Robot within 0.2%.

Item	Case	Hand	Robot	Diff.
Brace tension, bars 22/24 (bottom/top)	ULS+GY	5.44 / 5.66 kN	5.43 / 5.66 kN	0.2% / 0.0%
C1 axial force	ULS+GY	50.40 kN	50.40 kN	0.0%
Brace tension, bars 18/20	ULS+GX	2.41 / 2.63 kN	2.41 / 2.63 kN	0.0%
C1 axial force	ULS+GX	47.63 kN	47.63 kN	0.0%

ULS+GY is 1.25D + 1.5L + 1.5GY + 1.0NY, and it governs the post and brace design.

Displacements. Under SLS+GY (D + L + GY, unfactored), the drift at the post top was 0.30 mm (H/10 000), against 0.29 mm by hand. The lateral deflection of B1 at midspan was 4.65 mm (L/1290), against 4.67 mm by hand. The drift under SLS+GX was 0.15 mm.

CSA S16 member checks. The members were checked to CSA S16 in CAL-004, using the governing factored effects from the gravity and lateral load cases. All members pass, and the highest utilization is 0.53 in B1 under ULS+GY. In addition, the B1 result depends on lateral restraint from J1 (Finding 8). The weak-axis moment in B1 was taken as an upper bound, with the factored guard load of 1.125 kN/m carried by B1 alone as a simple span ($wL^2/8$), although in the frame it is shared with the joists and bracing.

Member	Check	S16 clause	Factored effect	Factored resistance	Util.
B1	Bending, Class 2, L_u = 1.5 m	13.6	M_f = 78.70 kN·m	M_r = 192.2 kN·m	0.41
B1	Shear	13.4.1.1	V_f = 35.96 kN	V_r = 373.8 kN	0.10
B1	Biaxial bending, ULS+GY	13.8.3	M_{fx} = 78.70, $M_{fy} \leq$ 5.06 kN·m	M_{rx} = 192.2, M_{ry} = 42.5 kN·m	0.53
J1	Bending, Class 4 flange, L_u = 3.0 m	13.5(c), 13.6	M_f = 13.51 kN·m	M_r = 44.3 kN·m	0.30
J1	Shear	13.4.1.1	V_f = 18.02 kN	V_r = 183.3 kN	0.10
C1	Compression, KL = 3.0 m, KL/r = 61.5	13.3.1	C_f = 50.40 kN	C_r = 661 kN	0.08
BR1	Tension, gross section yield	13.2	T_f = 5.66 kN	T_r = 241.9 kN	0.02
BR1	Net section, shear lag 0.60A_n	12.3.3.3, 13.2	T_f = 5.66 kN	T_r = 124.7 kN	0.05
BR1 bolt	Shear, threads intercepted	13.12.1.2	V_f = 5.66 kN	V_r = 87.1 kN	0.07
BR1 bolt	Bearing on 6.35 mm leg	13.12.1.2	B_f = 5.66 kN	B_r = 137 kN	0.04

In addition, the brace end connection was checked with one assumed M20 A325 bolt, and bolt shear was found to govern. The block shear, edge distances and end distances were not checked, since the connection geometry has not been fixed.

FINDINGS

The bracing was found to govern the review, since its slenderness requires a connection at the brace intersection and a tension-only arrangement. Each finding has been logged in the review register (AS-P1-MTO-001, REVIEW_REGISTER tab).

No.	Item	Finding	Action	Status
1	C1 reaction	Robot gave 46.83 kN against 46.03 kN by hand, since Robot includes the bracing self-weight (2.6 kN in total, about 0.8 kN per post at ULS).	Accepted as explained.	Closed
2	C1 member force	In R0 the Robot member force of 41.9 kN was lower than the hand value, since the bracing carried part of the gravity load.	Braces modelled as tension-only in MOD-001 R1; all braces inactive under gravity, and C1 = 46.55 kN against 46.03 kN by hand (+1.1%).	Closed
3	BR1 slenderness	With r_{min} = 12.5 mm (computed from I_{yp} and A), L/r is about 537 over the full length of the long braces, against limits of 300 in tension and 200 in compression.	Braces bolted at the intersection and treated as tension-only (S-101, S-201), and modelled as tension-only in MOD-001 R1. L/r = 268 (long) and 170 (short) at half length, below 300.	Closed
4	BR1 sag	A self-weight sag of about 24 mm (about L/280) was shown in Robot for the long braces.	Crossing bolt not modelled. In R1 the braces are truss members, so sag is not computed in Robot.	Open
5	Revit model	The schedule volume check showed the braces modelled as W6X15 (area about 2950 mm^2 against about 768 mm^2 expected).	Type corrected to the L64×64×6.4 equivalent and the schedule re-checked.	Closed
6	Material take-off	1298 kg of members was obtained from CISC mass per metre, against 1327 kg from the Revit volumes (+2.3%, imperial family geometry). An assumed 10% connection allowance gives 1428 kg in total.	Accepted, the allowance to be confirmed.	Closed
7	J1 flange class	b/2t = 11.5, which exceeds the Class 3 limit of $200/\sqrt{F_y}$ = 10.7, so the J1 flange is Class 4 in 350W.	Effective flange width used in CAL-004 (Clause 13.5(c)); utilization 0.30.	Closed
8	B1 lateral support	The B1 bending resistance assumes the compression flange is braced by J1 at 1.5 m. Unbraced over 6.0 m, the utilization would be 1.01.	Joist-to-beam connection to provide lateral restraint, to be confirmed in the connection design.	Open

CHANGES IN R1

1. Braces modelled as tension-only in MOD-001 R1 (bars 18–25). All braces are inactive under gravity, and Finding 2 is closed.
2. Notional loads (CSA S16 8.4.1) and a guard load of 0.75 kN/m (NBC 2020 4.1.5.14) added as lateral load cases. Brace and post forces and displacements checked by hand in CAL-003.
3. Live load and deflection criteria cited in the design basis (NBC 2020 Table 4.1.5.3; CSA S16 Annex D).
4. Member checks to CSA S16 added in CAL-004, including the brace end bolt (assumed 1 M20 A325; bolt shear and bearing). Other connections not designed.

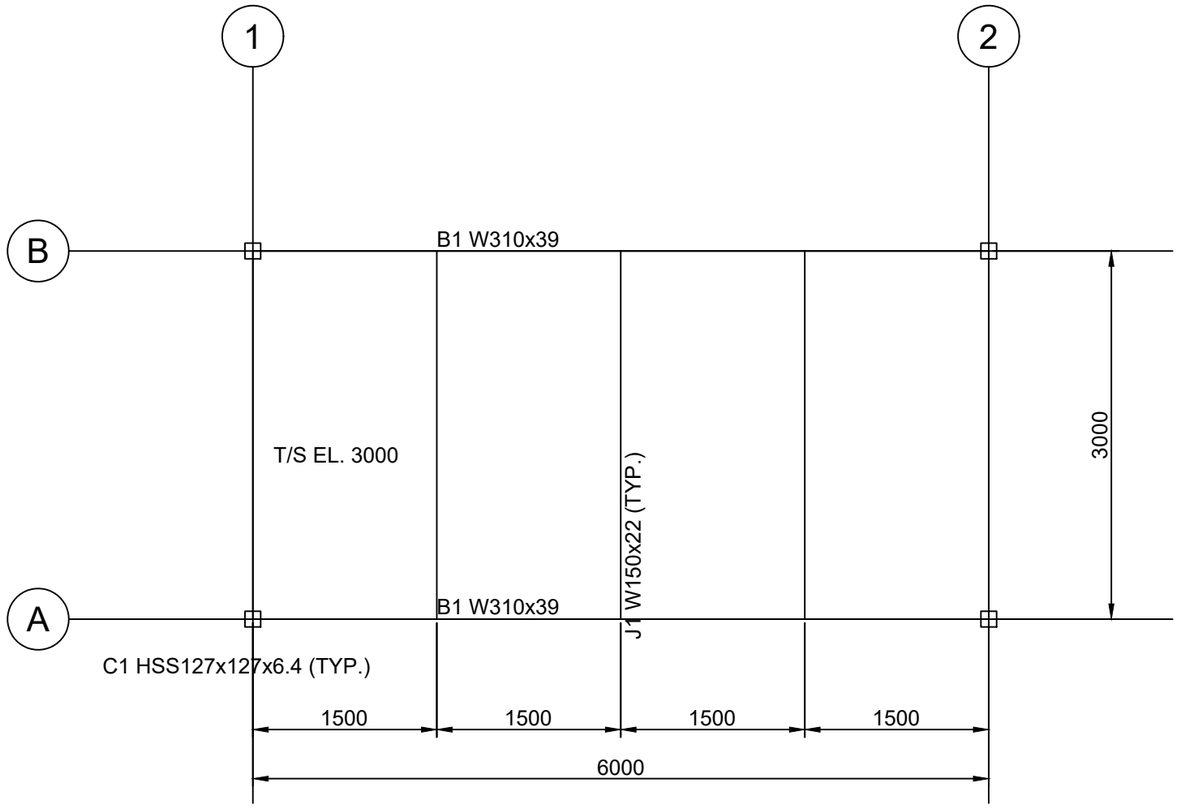
DOCUMENTS ISSUED

Document	Description	Rev	Document	Description	Rev
AS-P1-CAL-001	Mathcad hand calculation	R0	AS-P1-MOD-001	Robot 3D model and result plots	R1
AS-P1-CAL-002	Excel calculation and three-way check	R0	S-101, S-201	AutoCAD plan, sections, notes	R3
AS-P1-CAL-003	Hand check of lateral loads, brace forces and displacements	R1	AS-P1-MOD-002, S-301	Revit model, schedules, sheet	R0
AS-P1-CAL-004	CSA S16 member checks	R1	AS-P1-MTO-001	MTO, slenderness, review register	R1

Assumed 1 M20 A325 bolt per brace end, threads intercepted. Shear lag 0.60An (angle connected by one leg, fewer than 4 bolts). Bolt shear governs.

SUMMARY: CSA S16 MEMBER CHECKS	Factored	Resistance	Ratio	
B1 Bending, Lu = 1.5 m	78.70 kN·m	192.2 kN·m	0.41	
B1 Shear	35.96 kN	373.8 kN	0.10	
B1 Biaxial bending (ULS+GY)			0.53	
B1 Bending, Lu = 6.0 m (unbraced)	78.70 kN·m	77.7 kN·m	1.01	FAILS: needs J1 restraint
J1 Bending, Class 4, Lu = 3.0 m	13.51 kN·m	44.3 kN·m	0.30	
J1 Shear	18.02 kN	183.3 kN	0.10	
C1 Compression	50.40 kN	661.2 kN	0.08	
BR1 Tension, gross	5.66 kN	241.9 kN	0.02	
BR1 Tension, net	5.66 kN	124.7 kN	0.05	
BR1 Bolt shear (governs)	5.66 kN	87.1 kN	0.07	
BR1 Bolt bearing	5.66 kN	137.2 kN	0.04	

All members pass with J1 providing lateral restraint to B1.



X-BRACING BR1 L64x64x6.4 ALL FOUR SIDES, BOLTED AT INTERSECTION, TENSION-ONLY. SEE S-201

GUARD LOAD 0.75 kN/m (OBC 4.1.5.14) AND S16 NOTIONAL LOADS INCLUDED IN R1 ANALYSIS. SEE AS-P1-CAL-003.

FRAMING PLAN 1:50

MAINTENANCE PLATFORM VERIFICATION

S-101 FRAMING PLAN

SCALE 1:50 DRAWN A. SALEH 2026-10-17

STUDENT EXERCISE. NOT FOR CONSTRUCTION

REV 0 2026-10-17 ISSUED FOR INTERNAL CHECK

REV 1 2026-10-17 ISSUED FOR REVIEW

REV 2 2026-10-18 BR1 INTERSECTION NOTE ADDED

REV 3 2026-10-21 R1: TENSION-ONLY ANALYSIS, NOTIONAL AND GUARD LOADS

AS-P1-MTO-001 Material Take-Off, Maintenance Platform (Portfolio Study)

Rev R1, 2026-10-21. Lengths from Revit schedules (AS-P1-MOD-002, sheet S-301). Mass/m from CISC Handbook. Grey cells = inputs.

Mark	Member	Section (CISC)	Revit type	Qty	Length each (m)	Total length (m)	Mass/m (kg/m)	Mass (kg)	Notes
B1	Main beam	W310x39	W12X26	2	5.848	11.696	38.80	453.8	Revit cut length
J1 (end)	Joist	W150x22	W6X15	2	2.848	5.696	22.50	128.2	J1-1, J1-5
J1 (int.)	Joist	W150x22	W6X15	3	2.810	8.430	22.50	189.7	J1-2 to J1-4
C1	Post	HSS127x127x6.4	HSS5X5X1/4	4	3.000	12.000	23.20	278.4	System length GRADE to T/S
BR1 (long)	X-brace, long sides	L64x64x6.4	L2-1/2X2-1/2X1/4	4	6.463	25.852	6.01	155.4	BR1-1 to BR1-4
BR1 (short)	X-brace, short sides	L64x64x6.4	L2-1/2X2-1/2X1/4	4	3.848	15.392	6.01	92.5	BR1-5 to BR1-8
Subtotal, members				**19**		**79.066**		**1297.9**	
Connection allowance	Plates, bolts, welds						10%	129.8	ASSUMED % of member mass, not from code. Confirm with reviewer.
TOTAL STEEL								**1427.7**	**kg**

Project 2 – Conveyor Gallery Truss: Member Force Verification

Self-directed project | October 2026 | Rev 0, first issue

Tools: hand calculation, Robot Structural Analysis, Excel, AutoCAD

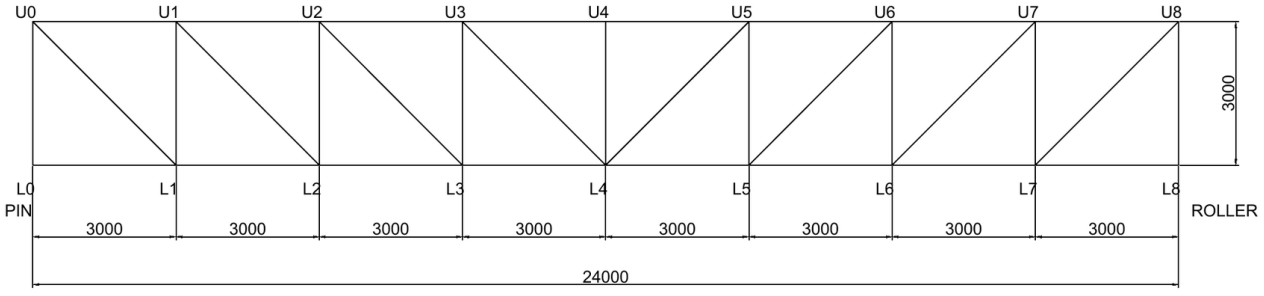
24 m	span, 8 panels × 3.0 m, 3.0 m deep
146.4 kN	maximum chord force (U3–U4, compression)
1,317 kg	steel take-off, one truss (1,449 kg with 10% connection allowance)

Description

Member-force verification of a 24 m, statically determinate Pratt truss for a conveyor gallery, with 8 panels at 3 m, 3 m deep, and pin and roller supports. Loads were worked out by hand to give reactions of 73.2 kN. Seven members were then hand-solved by the method of joints (L0, U0, L1) and a section cut through panel 4, and hand statics confirmed the Robot Structural Analysis member forces once Robot's sign convention (+ = compression) was converted. SLS deflection was checked, the elevation was drawn in AutoCAD (S-301) and a material take-off was prepared.

Key highlights

- Hand statics confirmed the Robot member forces for all 7 hand-solved members (AS-P2-CAL-001). The remaining 26 were taken from Robot and checked by symmetry.
- Governing chord force U3–U4 = 146.4 kN compression, from the section cut M/d = 439.2 / 3.0.
- Steel take-off is 1,317 kg per truss, or 1,449 kg with a 10% connection allowance.



CONVEYOR GALLERY TRUSS - ELEVATION 1:100

1. TRIAL SECTIONS: CHORDS HSS102x102x6.4; VERTICALS AND DIAGONALS L76x76x6.4. NOT DESIGN-CHECKED.
2. LOADS (ASSUMED): DEAD 2.0 kN/m, LIVE 2.4 kN/m. ULS 1.25D + 1.5L = 18.3 kN AT L1-L7.
3. MEMBER FORCES: SEE AS-P2-CAL-001 (HAND CHECK) AND AS-P2-CAL-002 (VERIFIED AGAINST ROBOT AS-P2-MOD-001).
4. CONNECTIONS NOT DESIGNED. STUDENT EXERCISE. NOT FOR CONSTRUCTION.

Figure 1: S-301 truss elevation, 1:100 (R0).

AS-P2-MTO-001 Material Take-Off: Conveyor Gallery Truss Verification

Prepared by: Abbas Saleh Date: 2026-10-18 Rev: 0 Based on S-301 Rev 0; trial sections; centreline lengths. Student exercise, not for construction.

Mark	Member	Section (CISC)	Qty	Length each (m)	Total length (m)	Mass/m (kg/m)	Mass (kg)	Notes
TC	Top chord	HSS102x102x6.4	8	3.000	24.00	18.20	436.8	U0-U1 to U7-U8
BC	Bottom chord	HSS102x102x6.4	8	3.000	24.00	18.20	436.8	L0-L1 to L7-L8
V	Vertical	L76x76x6.4	9	3.000	27.00	7.28	196.6	U0-L0 to U8-L8
D	Diagonal	L76x76x6.4	8	4.243	33.94	7.28	247.1	U0-L1 to U3-L4, U5-L4 to U8-L7
Subtotal, one truss			**33**		**108.94**		**1317.3**	
Connection allowance	Gusset plates, bolts, welds					10%	131.7	10% as in AS-P1-MTO-001; replace once connections designed
TOTAL, ONE TRUSS							**1449.0**	**kg**
TOTAL, TWO TRUSSES			**2**				**2898.0**	**kg (assumed one truss each side of gallery)**

Checks

Item	Value	Target	Status
Member count	33	33	OK - matches 2j - r = 33
Chord length, each chord (m)	24.00	24	OK - equals span
Vertical count	9	9	OK - 8 panels + 1
Diagonal length (m)	4.243	4.243	OK - 3 m panel x 3 m depth

Mass per metre = A x 7850 kg/m3, with A from CISC section properties (HSS102x102x6.4: 2320 mm2; L76x76x6.4: 927 mm2), same method as AS-P1-MTO-001. Lengths are node-to-node centreline lengths from S-301; no allowance for cut lengths or end preparation.

Maintenance Access in Hydropower Plants: Permanent Platforms and Scaffolding

AS-P3-RPT-001

Revision 0 18 October 2026

Literature Review Prepared by: Abbas Saleh

Research Question

For turbine maintenance which has to be done over and over again, what are the practical differences between using a permanent modular platform and setting up temporary scaffolding each time? This review looks at the fall hazards which come with each system, the amount of time which is needed to get access set up before the work can start, and the inspection and maintenance work which comes with each system.

Summary

From the sources which were reviewed, the main benefit of a permanent platform does not seem to come from the inspection itself. It comes from the work which has to happen before and after each inspection. The scaffold has to be built, checked, and then taken down again, and this is repeated every time access is needed. At the same time, the workers who are doing this are exposed to fall hazards while the scaffold is still incomplete. The sources which were found suggest that a permanent arrangement would be safer and would also take less time to get onto. However, no comparison was found where the hours saved or the incidents avoided were actually measured, and there was also very little information on what is needed to maintain a permanent platform over its life. Because of this, the case for a permanent platform at the moment is based on the reduction in repeated scaffold work rather than on any number which can be quoted.

1. Background

In many cases, inspecting the underside of a turbine runner is more of an access problem than an inspection problem. The work has to be done inside the draft tube, where there is not much room to move around. The usual approach is to build scaffolding inside the draft tube elbow so that the inspection areas can then be reached. After the work is finished, the scaffold is then removed. If access is needed again during a later outage, the same arrangement then has to be built again from the beginning.

Hatch (2026) published a different approach in January 2026 [1]. A modular access platform was developed for SCHM's McCormick Hydropower Plant in Québec. The project started in 2019 after a safety concern was found underneath the turbine runner during an inspection. From the information which Hatch provided, the finished system serves five turbine-generator units, works with three different turbine types, and removes the need for conventional scaffolding.

The reason this project was chosen is because turbine maintenance is repeated. A temporary solution may be fine when access is only needed once in a while. It becomes a lot less attractive when essentially the same working area has to be set up again every time the equipment is inspected. At the same time, this does not automatically mean that the permanent installation is the better option. It still has to be designed for the site, inspected, and also maintained. Because of this, what needs to be looked at is whether the amount of scaffold work which is avoided is actually worth more than the extra responsibilities which then come with the permanent platform.

2. Sources and Scope

Six publicly available sources were reviewed. These include Ontario safety legislation, provincial working-at-heights guidance, a University of Toronto scaffold standard, one peer-reviewed study, and project information which was published by Hatch and by ZARGES. The legislation and safety documents were used to find the requirements which apply to scaffolding and fall protection. The Hatch and ZARGES publications were then